# Jason Stack

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## Education

### Ph.D. in Mathematics — May 2027

- University of Texas at Dallas, Richardson, Tx. Advisor: Nathan Williams — 3.872 GPA
- Relevant Course Work: Stochastic Dynamic Programming, Machine Learning and Control Theory, Optimization, Nonlinear Analysis, Real Analysis II, Complex Analysis, Abstract Algebra II, Algebraic Topology.

### B.A. in Mathematics with honors, *magna cum laude* — Jun 2021

- Union College, Schenectady, NY. — 3.737 GPA

## Technical Skills

**Python · NumPy · Java · Mathematica · MATLAB · SageMath · Gap3 · LaTeX**
**Canvas ·Blackboard**

## Employment

### Residential Assistant, Union College, Schenectady, NY — Sep 2018 – Jun 2019

- Cultivate a welcoming, inclusive, environment for first year students. Work as a liaison to bridge the gap from students to administrators to build a more student friendly experience across campus.

### PhD Student, University of Texas at Dallas, Richardson, Tx — Aug 2022 – Present

- Assisted in writing and editing a grant proposal to the NSF with Nathan Williams, resulting in a 36 month long grant awarded for the amount of $210,000.
- Written and received applications for travel awards given through the School of Natural Sciences and Mathematics, and the Office of Graduate Education, as well as multiple conference specific travel awards. Managed award funds through reimbursement filings and interdepartmental transfers. Total amount in travel awards received: $3,200.

## Teaching and Mentoring

### Teaching Assistant, University of Texas at Dallas, Richardson, TX — Aug 2022 – Present

- Created and led lecture sections for various math courses, and graded homework, quizzes, and exams. Communicated with professors and met consistently with students to improve student outcomes.

### Teaching Assistant, Tulane University, New Orleans, LA — Aug 2021 – May 2022

- Assisted mathematics professors with grading of homework, tests, and quizzes in calculus I and statistics. Collaborated with professors and fellow teaching assistants to create lectures and quizzes.

### Calculus Tutor/Counselor, Union College, Schenectady, NY — Jun 2019; Jun 2020

- Developed curriculum, created lectures, and tutored to prepared incoming first year students in the Academic Outreach Program, increasing prospects for students and graduate rates in the program.

### Mathematics Assistant, Union College, Schenectady, NY — September 2017 – 2021

- Collaborated with mathematics professors in variety of courses developing lessons, grading, and tutoring students improving student understanding and metrics.

### Undergraduate Mentor, University of Texas at Dallas, Richardson, Tx — Aug 2022 – Present

- Developed research expertise amongst undergraduates, resulting in continued education at the PhD level for two of the mentored students, one in mathematics and and one in computer science. Porject resulted in the published paper *An elbaorate new proof of Cayley's formula.*
- Assisted with Putnam club. Fostered mathematical talent amongst undergrads, resulted in UTD being ranked 42nd out of 487 universities, with two students ranked in the top 500 in 2026.

## Service

**NSM Ambassador, University of Texas at Dallas.** Dallas, TX 2026 – Present

Represented NSM touring incoming students, discussing graduate programs with current undergraduates, and assisting administrators in career center or outreach programs. Volunteered math tutoring through Top Achievers.

**Tutor, Girls Inc.** Schenectady, NY 2017 – 2021

Assisted elementary school girls with homework after school twice a week. Worked alongside permanent employees to foster a welcoming and productive learning atmosphere.

## Research Experience

**PhD Research, University of Texas at Dallas,** Dallas, TX August 2023 – Present

- Conducted research extending results for real reflection groups to complex reflection groups.
- Filled a gap in the literature and exposed a variety of avenues for future research topics. Resulted in a paper submitted for publication, *Parking Spaces for Complex Reflection Groups*, and further works in progress.

**Research Assistant, University of Texas at Dallas,** Dallas, TX August 2025 – May 2026

- Leveraged computation software to explore potential research problems. Extended known results to a more general setting.
- Answered a known open question that resulted in bridging a research gap in algebraic combinatorics.

**Thesis, Union College,** Schenectady, NY September 2020 – June 2021

- Under the advisement of Professor Leila Khatami, continuing research into non-similar multiplication maps in an Artinian algebra $\mathbf{R}$.
- Utilized Python and SageMath to develop a diagram to represent Hilbert function and Jordan types in different Artinian algebras of any dimension. This research contributed to extending a known 2-dimensional model to any dimension, broadening the ability to leverage these mathematical tools in more complex settings.

**Independent Researcher, Union College,** Schenectady, NY September 2020 – June 2021

- Collaborating with mathematics Professor Jeffrey Hatley studying elliptic curve ranks, of particular interest in quantum computing and cryptography.
- Leveraging SageMath, discovered certain families of elliptic curves and providing further evidence for conjectured results in the field. This work ultimately resulted in the publication *Two infinite families of elliptic curves of rank greater than one.*

**Research Assistant, Union College,** Schenectady, NY June 2020 – August 2020

- Analyzed elliptic curve congruence versus $a_l$ values using SageMath alongside Professor Jeff Hatley. Acquired data that formed the basis for further research inquiries.

**Research Assistant, Union College,** Schenectady, NY June 2020 – August 2020

- Studied non-similar multiplication maps in an Artinian algebra using Python and SageMath with Professor Leila Khatami. Developed individual mathematical research understanding and skills.

**Research Assistant, Union College,** Schenectady, NY June 2018 – August 2018

- Built upon previous student work under Professor Chad Orzel to explore the attration of lasers on particles under a mirror setup. Improved individual research ability and developed techniques for future students to leverage in related research.

## Journal Publications

E. Banaian, A. Hoang, E. Kelley, W. Miller, **J. Stack**, C. Stephen, and N. Williams. *An elaborate new proof of Cayley's formula.* **Algebraic Combinatorics**, 8.4 (2025): 971-995.

J. Hatley and **J. Stack**. *Two infinite families of elliptic curves of rank greater than one.* **Integers**, 22, #A1 (2022), 15pp.

## Preprints and Works in Progress

2026 **J. Stack**, and N. Williams. *Full-Support Parking Spaces.*

2026 **J. Stack**, and N. Williams. *Posh Parking Spaces.*

2025 **J. Stack**. *Parking spaces for complex reflection groups.* arXiv:2502.01970.

## Refereed Conference Publications

**J. Stack**. *Parking spaces for complex reflection groups.* Proceedings of the 38th Annual Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2026), Seattle, Washington.

## AMS Invited Presentations

2025 *Parking spaces for complex reflection groups*, AMS Fall Section Meeting, St. Louis University.

## Presentations

2026 *Positive noncrossing parking spaces*, CombinaTexas, Texas A&M.

2025 *Noncrossing parking functions*, SMU-UTD SIAM student chapter, University of Texas at Dallas.

2025 *Parking spaces for complex reflection groups*, CombinaTexas, Texas A&M.

2025 *Parking spaces for complex reflection groups*, UTD Research Days, University of Texas at Dallas.

2024 *Parking spaces for complex reflection groups*, External Advisory Council, University of Texas at Dallas.

2024 *Noncrossing parking functions and complex reflection groups*, UTD Research Days, University of Texas at Dallas.

2021 *All Jordan Type Partitions for a Hilbert Function of the Complete Intersection Type*, Steinmetz Day, Union College.

## Awards and Memberships

Union College Scholar (2017-2021)

Dean's List (2017-2021)

Elected to Pi Mu Epsilon Mathematics Honor Society Member (2020)

American Mathematics Society membership granted by Union College mathematics department (2020)

Society for Industrial and Applied Mathematics membership (2021)

CombinaTexas Travel Award, $150 (2023,2024,2025)

NSM Travel Award, $750 (2025)

NSM Summer Research Fellowship, (2025)

Betty and Gifford Johnson Travel Award, $2,000 (2026)